

Zain Mahmoud

AI & Software Developer | 3B Mechatronics Student at the University of Waterloo

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🌐 [GitHub](#)

🌐 [LinkedIn](#)

🌐 [Website](#)

Skills

- **Languages & Platforms:** Python, JavaScript/TypeScript, Rails, C++, SQL, Java, Scala, Kotlin, cross-platform (Linux/Windows/macOS)
- **Backend & Data:** Supabase/PostgreSQL (schema design, indexing, RLS), low-latency REST APIs, async pipelines, distributed systems patterns, background task queues, caching, rate-limiting, load-balanced microservices
- **Frameworks & Servers:** FastAPI, Flask, Node.js, Express, Uvicorn, Rails, Next.js (API routes), Nginx
- **AI Infrastructure:** OpenAI, Hugging Face; built production RAG systems with Qdrant/FAISS, embedding pipelines, and CUDA-accelerated compute
- **Cloud & DevOps:** Docker, GCP, AWS, CI/CD with GitHub Actions, HTTPS load balancing, secret management
- **APIs & Integrations:** Stripe (wallets/payments), GraphQL, ADB automation, RQ task queues, CUPS, ngrok tunneling
- **Systems & Architecture:** Token-based auth, WebSockets/SSE, microservices, real-time AI interfaces, scalable backend design

Relevant Experience

Forward Deployed Developer Co-op

Aug 2025 – Present

Fuego Studios Inc, San Francisco, CA ([link](#))

- Built high-impact React Native components using the Fuego SDK, powering Shopify merchants' mobile storefronts.
- Implemented GraphQL product/variant queries, dynamic pricing, carts, and native storefront sections.
- Created theme-ready UI modules reused across multiple merchant apps, improving build velocity and consistency.
- Optimized performance with batched GQL requests, local caching, and faster render paths on product pages.
- Debugged production storefronts, fixed schema mismatches, and ensured stable end-to-end API integration.

AI & Full Stack Developer Co-op – Jilani Place App

Jan 2025 – April 2025

Jilani Place Inc, Etobicoke ON ([link](#))

- Replaced a legacy SaaS with a custom full-stack coworking app using React Native (Expo), FastAPI, and Supabase, cutting monthly software costs by 90%.
- Built an AI assistant with Qdrant (RAG), OpenAI, and Hugging Face that automated 80% of booking/support queries, reducing response time from hours to seconds.
- Delivered role-based dashboards and Stripe Wallet payments, reducing friction and enabling full self-service.
- Automated Android room displays and print jobs with RQ, ADB, and CUPS for real-time facility control.

AI & Software Developer Co-op – AI Healthcare Planner

Feb 2024 – December 2024

Workhorse Health, Toronto ON ([link](#))

- Built an AI healthcare planning platform with React (Next.js), Node.js, FastAPI, and Firebase for real-time interactive workflows.
- Integrated OpenAI, Langchain, Hugging Face, and CUDA for fast, GPU-accelerated summarization, cutting staff workload by up to 60x.
- Built a medical RAG system for accurate, scalable clinical answers and automated data retrieval.
- Developed a Selenium + BeautifulSoup scraper refined by an LLM to embed trusted medical content into the AI database.
- Deployed the containerized app to Google VMs using a cloud registry and load balancer for production uptime.

Education

University of Waterloo, Waterloo, ON

2022 - 2027

Bachelor Of Applied Science (BASC) In Mechatronics Engineering Honors, Co-operative Program

Relevant Coursework: Digital Computation, Algorithms and Data Structures, Intro to Microprocessors and Digital Logic (GPA: 3.0/4.0)

George Brown College, Toronto, ON

2019 - 2022

Advanced Diploma in Electromechanical Engineering Technology (GPA: 3.8/4.0; Dean's List)

Personal Projects

AI-Powered Construction Estimator ([link](#))

January 2025 – Present

- Built an AI construction planner with React Native, Supabase, Qdrant, and fine-tuned GPT models that generates full build plans, sources materials with real-time Home Depot API calls, and delivers role-based flows for Residential, and Contractor/Worker users.

Portfolio Website ([link](#))

May 2024 – July 2024

- Developed a responsive React + Tailwind portfolio with a Node.js/Supabase backend and an integrated Llama-powered AI guide for interactive browsing and improved user experience.